MIL-C-10405/8A(EL)
18 September 1967

SUPERSEDING
MIL-C-10405/8(SigC)
2 April 1954

MILITARY SPECIFICATION

CRYSTAL UNIT CR-13/U

1. SCOPE

1.1 This specification covers the detail requirements for the crystal unit CR-13/U. This crystal unit has a quartz vibrator spacer mounted in a metal holder, type HC-17/U, and is designed to operate at series resonance, on the fundamental frequency of the vibrator, under noncontrolled temperature conditions, at a specified frequency of 455 kilohertz (KHz).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-10405 - Crystal Units, Quartz, Pressure and Spacer Mounted.
MIL-H-10056/8 - Holder, Crystal, HC-17/U.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

MILITARY

SC-D-17499 - Marking Instructions for Crystal Units.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5955

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Table 1. Detail requirements

Requirements	Specified values or conditions
Holder	HC-17/U (per MIL-H-10056/8
Dimensions	(see MIL-H-10056/8)
Frequency range	455 kilohertz, exclusive
Frequency tolerance	±0.02 percent
Mode of oscillation	Fundamental
Temperature range	-40° to + 70°C, inclusive
Shunt capacitance	14 ± .5 pf (holder capacitance)
Drive level	The crystal drive control of the test set shall be set for the minimum value at which the crystal will oscillate.
Maximum resonance resistance	5,000 ohms
Government reference standard test set	TS-537/TSM
Springs	Coil type; Minimum force 2½ lbs,(see 3.5.1 of MIL-C-10405)
Marking	Per drawing SC-D-17499; method 1.
Shock:	
Frequency change permitted	0.002 percent
	The resistance shall not exceed 5,000 ohms
Vibration: 1/	
Frequency change permitted	0.002 percent
	The resistance shall not exceed 5,000 ohms

1/ Method 201 of MIL-STD-202, amplitude 0.06 inch total excursion.

3. REQUIREMENTS

3.1 Requirements shall be in accordance with MIL-C-10405 and as specified herein.

3.2 The value of C in the series arm of the equivalent network of the crystal unit shall fall between 0.004 and 0.008 pf, when computed using the following equation:

$$C = \frac{2f\,(Co + C1)}{f_s}$$

Where f = the difference in frequency in cps between the series resonance and antiresonance frequency.

f_s = nominal crystal frequency in cps at series resonance.

Co = holder capacitance

C1 = capacitance which is put in series with the crystal unit when the series resonance-antiresonance switch is set at S. (See test set

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instructions for values of C1 to be used with above equation). C1 and Co shall be determined at room temperature.

The value of Co shall be measured at room temperature at a frequency below the fundamental frequency of the crystal plate and at which the crystal shows no resonant response.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Sampling and inspection shall be in accordance with MIL-C-10405 and as specified herein.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-C-10405.

6. NOTES

6.1 The notes specified in MIL-C-10405 are applicable to this specification.

CUSTODIAN:
ARMY - EL

PREPARING ACTIVITY
ARMY - EL

(Project 5955-A049)

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